MIL-E-18886(NOrd)

30 June 1955

SUPERSEDING
NAVORD OS 3065A
17 March 1945

MILITARY SPECIFICATION

EXTENDER, BOOSTER, DEPTH CHARGE, MARK 6 MOD 2

1. SCOPE

1.1 This specification governs the manufacture, assembly, and preparation for delivery of the Depth Charge Booster Extender Mechanism Mk 6 Mod 2 and the methods of inspection and test upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings, and publications of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

NN-B-621b	-- Boxes; Wood, Nailed and Lock-Corner
PPP-B-601	-- Boxes; Wood, Cleated-Plywood, for Domestic Shipment.

MILITARY

MIL-C-15167	-- Rust-preventative, thick film, (pigmented and nonpigmented).
MIL-P-116	-- Preservation, Methods of

STANDARDS

MIL-STD-105	-- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-- Marking for Shipments and Storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:03Z
Check the source to verify that this is the current version before use.

DRAWINGS

SK 109322 -- Depth Charge Booster Extender Mechanism Mk 6 Mod 2 and all specifications, standards, and drawings listed thereon.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.

3.1.1 The depth charge booster extender mechanism covered by this specification shall be manufactured in accordance with SK 109322 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Springs. -- All springs shall be made in accordance with the information given in the drawings. Particular care shall be exercised during the rust-proofing processes in order that the physical properties of the springs may not be adversely affected.

3.3 Rust-preventive compound. -- All metal parts shall be coated with rust-preventive compound in accordance with Military Specification MIL-C-15167. This coating shall be applied by dipping or spraying and surplus liquid shall be drained or blown off with compressed air.

3.4 Finished parts. -- All parts comprising the booster extender shall be manufactured in strict accordance with the drawings and within the tolerances specified thereon, both for individual parts and for subassemblies where indicated. Particular care shall be exercised at all times to prevent damage to the metallic bellows.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:03Z
Check the source to verify that this is the current version before use.

3.5 Strength of soldered joints and watertightness.-- The booster extender hydrostatic unit subassembly, Drawing 433983, shall comply with the requirements specified in 4.4.1, when air pressure is applied to the interior through suitable fittings engaging the booster carrying flange.

3.6 Spindle locking gear. -- The booster extender mechanism shall be tested for operation of the spindle locking gear as specified in 4.4.2.

3.7 Pressure requirements. -- The booster-extender shall be capable of withstanding the pressure tests specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the government inspector in the determination of the acceptability of products submitted by a supplier for government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Lot. -- As applied to the government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of five (5) complete depth charge booster extender mechanisms including both plain and knobbed safety forks accompanied by two (2) spare booster extender mechanism bellows shall be delivered by the contractor for test to an agency designated by the procuring agency. These depth charge booster extender mechanisms, safety forks and bellows shall be manufactured by the same procedures and processes and at the same location proposed by the contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the contractor will be notified and will be authorized by the procuring agency to proceed with the submission of the product for acceptance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:03Z
Check the source to verify that this is the current version before use.

Accepted samples will become the property of the procuring agency and will be included in the number called for in the contract schedule. The preproduction sample shall be considered satisfactory if for tests and inspections specified, it meets the acceptance criteria of MIL-STD-105, code letter C, AQL 4.0%.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Inspection. -- The general inspection procedures shall be in accordance with General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP 400; government inspection shall be in accordance with the provisions of General Procedure for Government Inspection and contractor's inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.4 Test procedures.

4.4.1 Strength of soldered joints and watertightness. --

4.4.1.1 The soldered joints and bellows shall show no evidence of leakage or failure under a pressure of 125 p.s.i.

4.4.1.2 There shall be no evidence of leakage under a pressure of 50 p.s.i. when the exterior surfaces and soldered joints are painted with a soap suds solution or, as an alternative, when the subassembly is immersed in water.

4.4.1.3 For tests specified in 4.4.1.1 and 4.4.1.2 the unit subassembly shall be secured in the test fixture in such a manner as to prevent longitudinal movement of the bellows and the 7/16-inch tapped hole in the hydrostatic piston shall be closed with a suitable plug. After these tests all parts shall be thoroughly cleaned and dried.

4.4.2 Spindle locking gear. -- With the assembled booster extender mechanism mounted so that the longitudinal movement of the hydrostatic piston and booster spindle is limited to 1.25 inch, remove the safety fork. The booster spindle should then move forward approximately 0.25 inch, opening a passage for the entrance of air around the end of the booster spindle. Air pressure

Source: https://assist.dla.mil -- Downloaded: 202640-18T12:03Z
Check the source to verify that this is the currentversion before use.

of between 6.0 and 11.0 p.s.i. applied to this opening should cause the hydrostatic piston and booster spindle to move forward the full allowed travel of 1.25 inch. The actural pressure at which unlocking occurs should be noted and recorded. Precaution must be taken to see that the packing gland in the spindle bushing is properly made up.

4.4.3 Pressure tests.

4.4.3.1 With safety fork in place and packing gland in spindle bushing adjusted to tightness suitable for operation as specified in 4.4.2 apply external water pressure of 20 p.s.i. Leakage of water around spindle bushing gasket or into the interior of the booster extender shall not be more than 10 cc. after this pressure has been applied for a period of one minute.

4.4.3.2 With the safety fork removed, apply hydraulic pressure of 400 p.s.i. No evidence of leakage shall be present under this condition. Increase hydraulic pressure to 500 p.s.i. and hold for a period of approximately 30 seconds. Leakage, if any, under this condition shall not be in excess of 5 cc. of fluid.

4.4.3.3 If the booster extender mechanism tested with pressure tests as specified in 4.4.3.1 and 4.4.3.2 fails to meet either or both requirements, acceptance of the entire lot shall be withheld pending an investigation of the failure. The booster extender mechanism shall be disassembled to determine the cause of the failure. Upon determination of the cause of the failure, suitable action shall be taken by the contractor to correct the lot for the defect prior to resubmission of the lot for acceptance. The particular test on which failure occurred shall be repeated and found satisfactory prior to acceptance of the resubmitted lot.

4.4.3.4 Following the maximum pressure test, the booster extender shall be disassembled and all parts examined. The booster extender used in the maximum pressure test shall be classified acceptable as part of a contract lot, provided that any damaged parts are replaced, and that the routine tests of water-tightness and of the spindle locking gear are subsequently applied.

Source: https://assist.dla.mil -- Downloaded: 202610-18T12:03Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -- Each extender mechanism shall be individually packaged in accordance with Drawings 421162 and 421163.

5.1.1 In addition, each unit package prepared as specified in 5.1 shall be packaged in accordance with Method IIb (less desiccant) as specified in MIL-P-116.

5.2 Packing. -- Ten (10) unit containers packaged as specified in 5.1 and 5.1.1 shall be packed in accordance with Drawing 443986.

5.2.1 Length, width, and depth of the shipping container must be modified to accommodate Method IIb packaging of the unit container.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in accordance with MIL-STD-129, "Marking for Shipment and Storage."

6. NOTES. -- There are none applicable.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 202610-18T12:03Z
Check the source to verify that this is the current version before use.